BOUNDARY LAYER CFD | OPENFOAM 13

# Surface Radiation Cuts Flat-Plate Wall Temperature by 6.98 K

Coupled convection-radiation CFD in OpenFOAM 13, validated against Blasius similarity

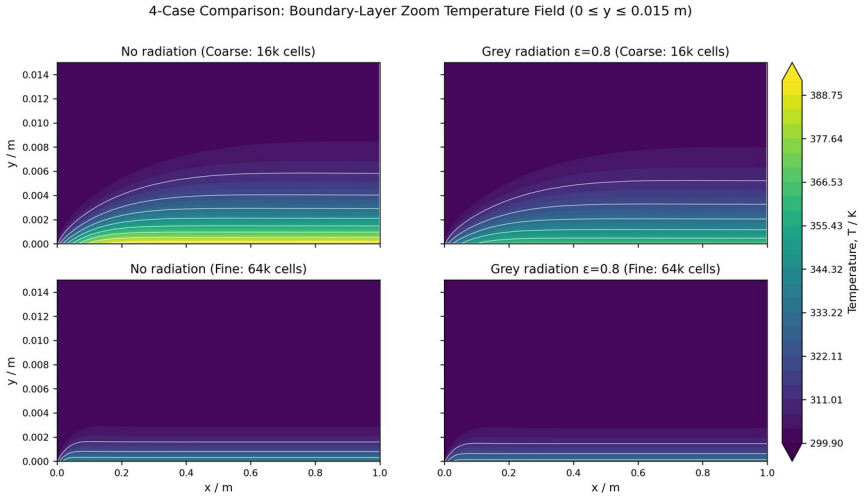



Fig. 1: Near-wall boundary layer temperature contours across test matrix.

Jerlin S. | Boundary Layer CFD Research | October 2026

# Neglecting Surface Radiation Overpredicts Heated Wall Temperatures

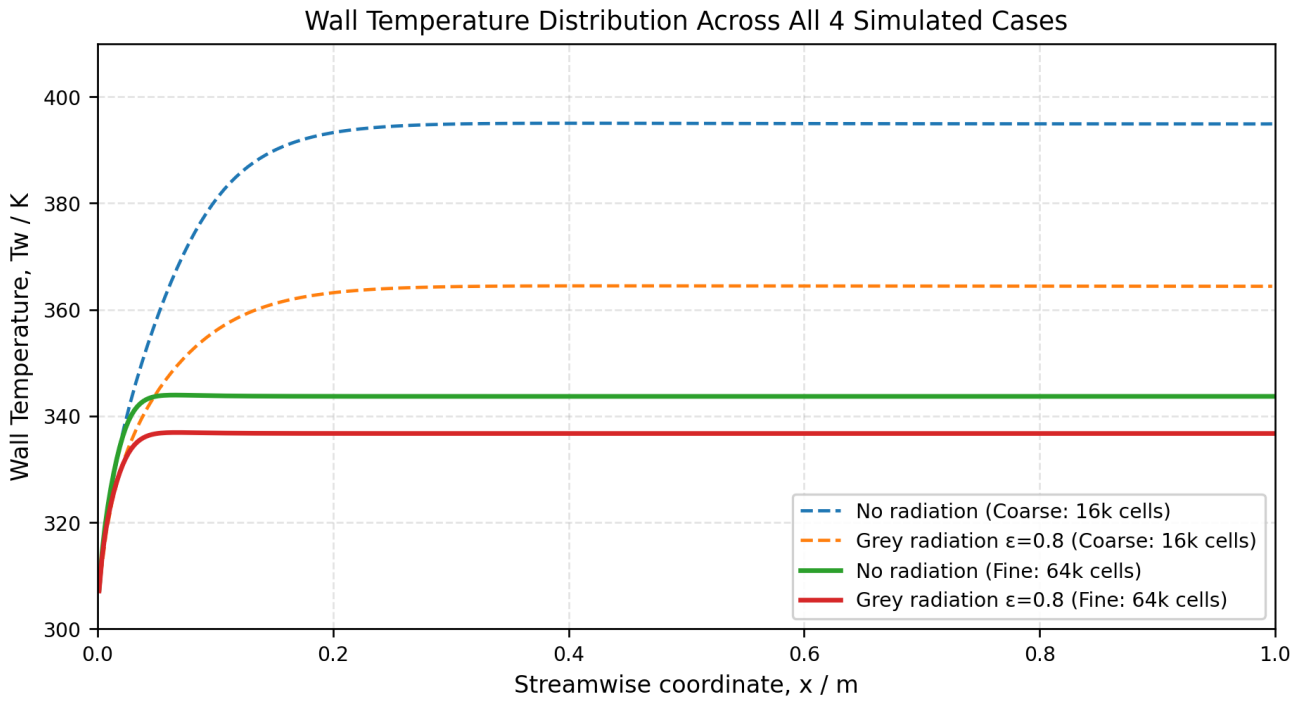



Fig. 2: Streamwise wall temperature distribution along heated flat plate.

- **Plate wall temperature depends strictly on competing heat rejection pathways.**
- Convective cooling drops downstream as hydrodynamic boundary layers thicken.
- Standard CFD workflows frequently omit radiation, overpredicting component peak temperatures.

**Heater Heat Flux: $q'' = 1000$ W/m² (uniform)**

# Four-Case Computational Matrix Isolates Radiation and Grid Effects

## 2 × 2 Factorial Simulation Matrix

| Case Identifier | Grid (nx×ny) | Cells | Emissivity ε | Heater q″ |
|---|---|---|---|---|
| noRadiation_coarse | 200 × 80 | 16,000 | 0.0 (Pure) | 1000 W/m² |
| noRadiation_fine | 400 × 160 | 64,000 | 0.0 (Pure) | 1000 W/m² |
| surfaceRadiation_coarse | 200 × 80 | 16,000 | 0.8 (Coupled) | 1000 W/m² |
| surfaceRadiation_fine | 400 × 160 | 64,000 | 0.8 (Coupled) | 1000 W/m² |

- Plate length L = 1.0 m, free-stream velocity U_inf = 2.0 m/s.
- Ambient and enclosure temperatures held constant at 300 K.
- Reynolds number Re_L = 1.258 × $10^5$ preserves laminar flow regime.

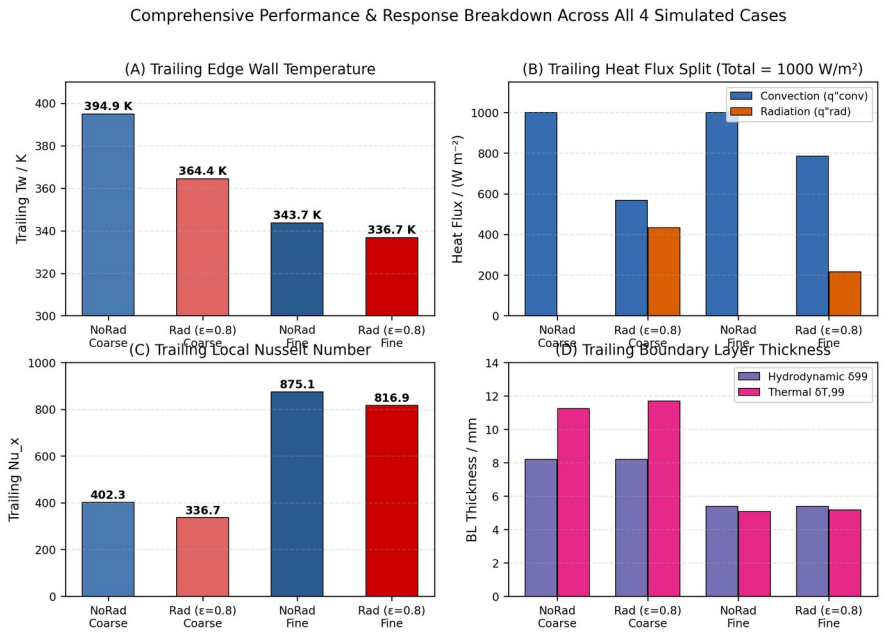



Fig. 3: Comprehensive parameter breakdown across all four cases.

# 80:1 Wall Grading Resolves Viscous and Conductive Sublayers

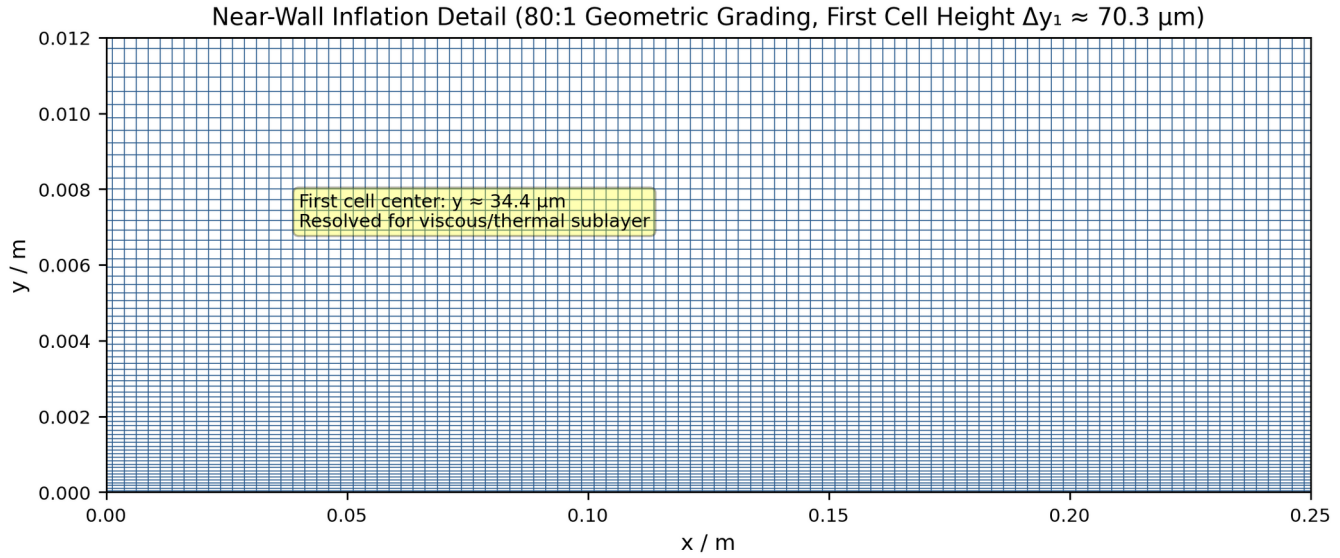



Fig. 4: Structured inflation mesh near heated plate leading edge (80:1 grading).

**41 Cells in 5 mm**

Near-wall inflation zone resolves thermal gradients

**$\Delta y_1$ = 70.3 μm**

First cell wall-normal height on 64k fine grid

- Domain includes 1.0 m unheated inflow section to develop boundary layer.
- Geometric wall-normal grading of 80:1 concentrates 41 cells in 5 mm.
- Fine 64,000-cell mesh maintains zero non-orthogonality and skewness.

# Newton-Raphson Solver Enforces Non-Linear Surface Heat Balance

## Wall Boundary Energy Conservation

**1. Surface Heat Balance (enforced at wall face):**

$$q''_{heater} = q''_{conv} + q''_{rad} = 1000\ W/m^2$$

**2. Stefan-Boltzmann Surface Radiation:**

$$q''_{rad} = \varepsilon\ \sigma\ (T_w^4 - T_{sur}^4)$$

where $\varepsilon = 0.8$, $\sigma = 5.670 \times 10^{-8}\ W/(m^2 \cdot K^4)$, $T_{sur} = 300\ K$

**3. In-Situ Newton-Raphson Iteration:**

$$T_w^{(k+1)} = T_w^{(k)} - f(T_w^{(k)}) / f'(T_w^{(k)})$$

- **Solves non-linear quartic radiative flux at every boundary face.**
- Enforces local face heat balance by construction during each PIMPLE step.
- Newton-Raphson scheme converges to double precision within four iterations.

**ε = 0.8**

Grey-diffuse emissivity to 300 K external enclosure

**< 4 Iter**

Fast quadratic convergence per boundary face

# Residuals Fall by 5+ Orders Across 1,500 Iterations

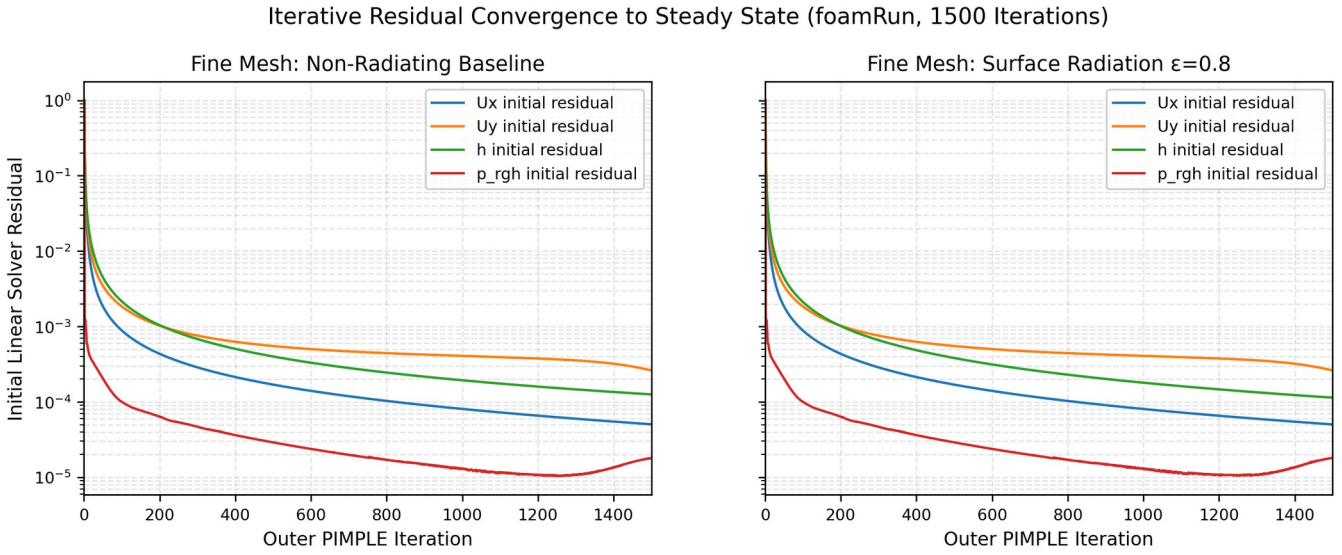



Fig. 5: Iterative residual history for velocity, pressure, and enthalpy.

- **Solved with OpenFOAM 13 foamRun using steady-state PIMPLE algorithm.**
- Residuals fall by 5+ orders, reaching monotonic asymptotic plateau.
- Constant properties decouple momentum from temperature under zero gravity.

**1,500 Iterations Completed**

**> $10^{-5}$ Residual Reduction**

**0.00 Max Non-Orthogonality**

# Surface Radiation Carries 21.3% of Total Heat Along Plate

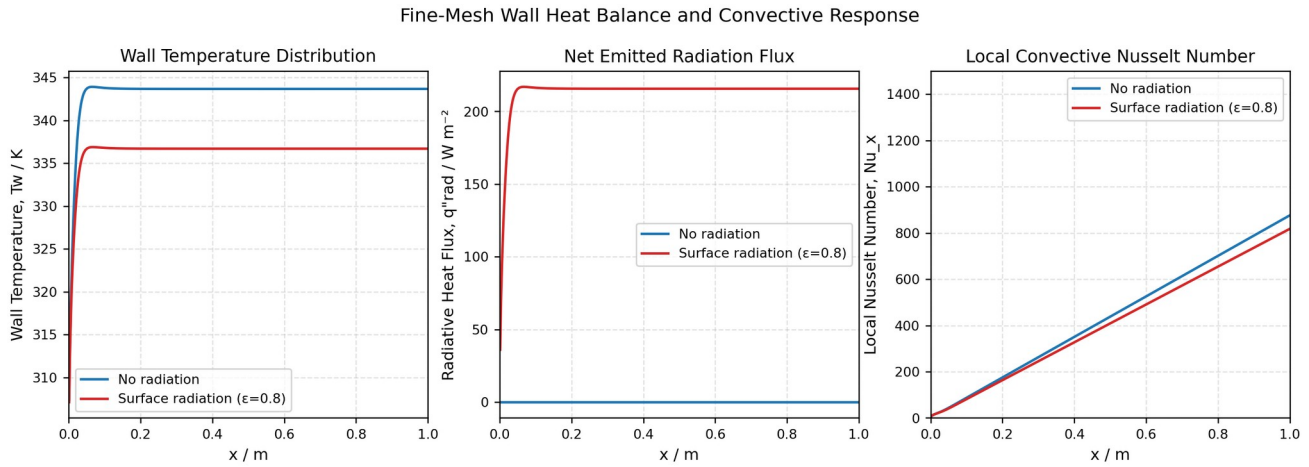



Fig. 6: Streamwise convective vs radiative heat-flux split and Nusselt number.

**21.3%**

**Plate-Averaged Radiative Load**

Mean flux: 213.2 W/m² (preliminary, fine-grid)

**21.6%**

**Trailing-Edge Radiative Load**

Trailing flux: 215.6 W/m² at x = 1.0 m

- Surface radiation relieves over one-fifth of the total heat dissipation load.
- Radiative heat flux remains nearly uniform along the entire plate surface.
- Convective flux drops downstream as thermal boundary layer growth increases resistance.

# Surface Radiation Depresses Peak Wall Temperature by 6.98 K

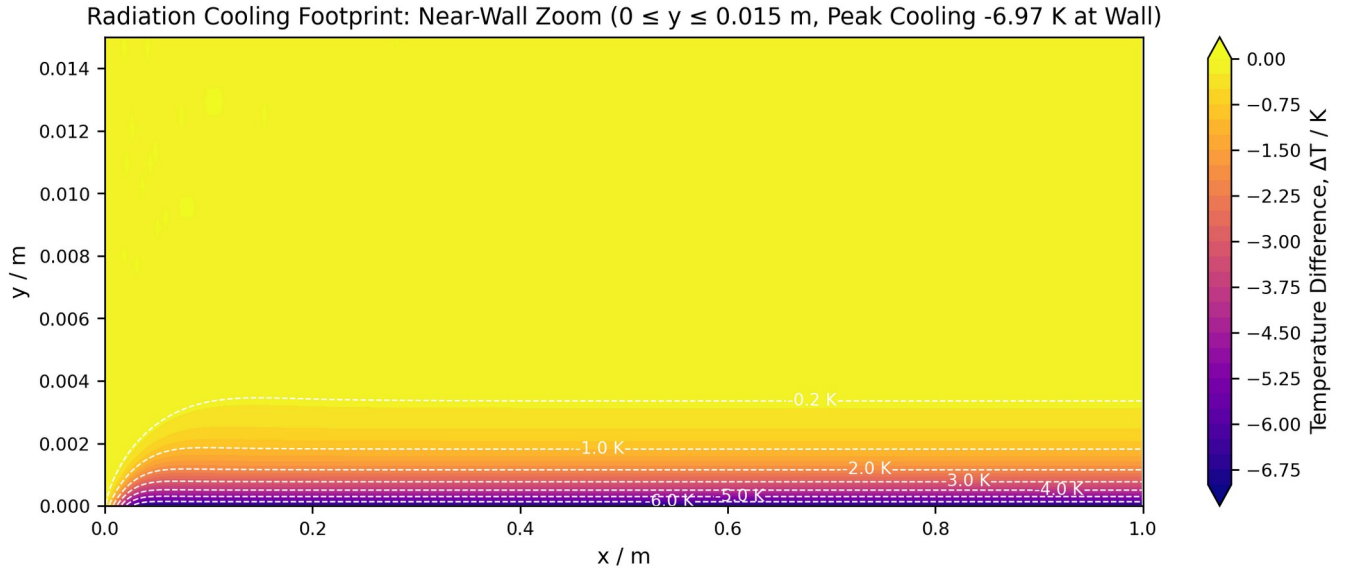



Fig. 7: Boundary layer temperature difference field (T_radiation - T_baseline).

## 6.98 K Drop

**Peak Wall Temperature Reduction (Fine Grid)**

- Radiation lowers plate wall temperature along the entire heated length.
- Cooling effect intensifies downstream, reaching maximum depression of 6.98 K.
- Thermal boundary layer thickness remains virtually unchanged across radiation cases.

Fine grid trailing edge: 336.70 K vs 343.68 K pure convection.

# Skin Friction Validates Against Blasius Within 0.11%

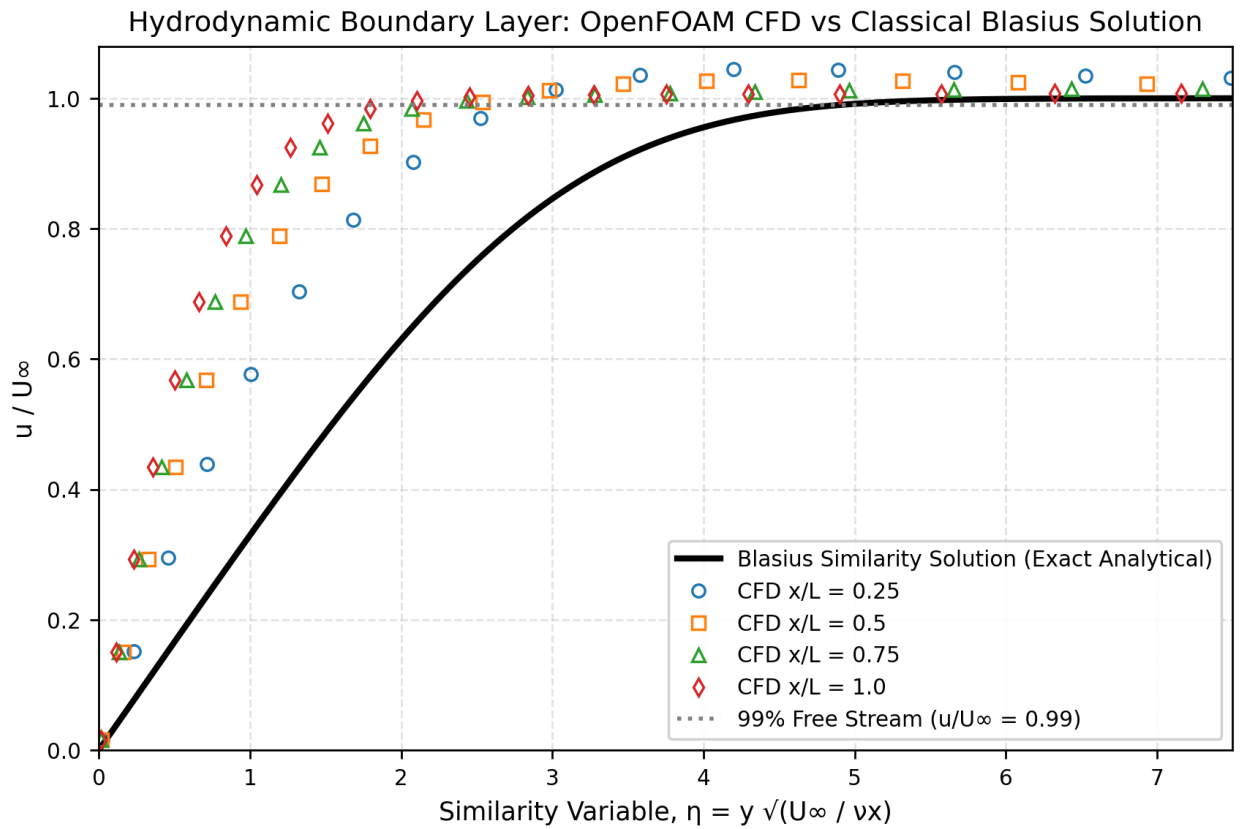



Fig. 8: Non-dimensional streamwise velocity collapse vs exact Blasius curve.

**0.11%**

Skin-friction relative error vs Blasius (Cf = 1.874e-3 vs 1.872e-3)

**$1.26 \times 10^5$**

Plate Reynolds number (Re_L = U_inf L / ν) confirms laminar flow

**Invariant**

Velocity profiles strictly identical between radiating and baseline cases

# Thermal Boundary Layer Tracks Self-Similar Growth Dynamics

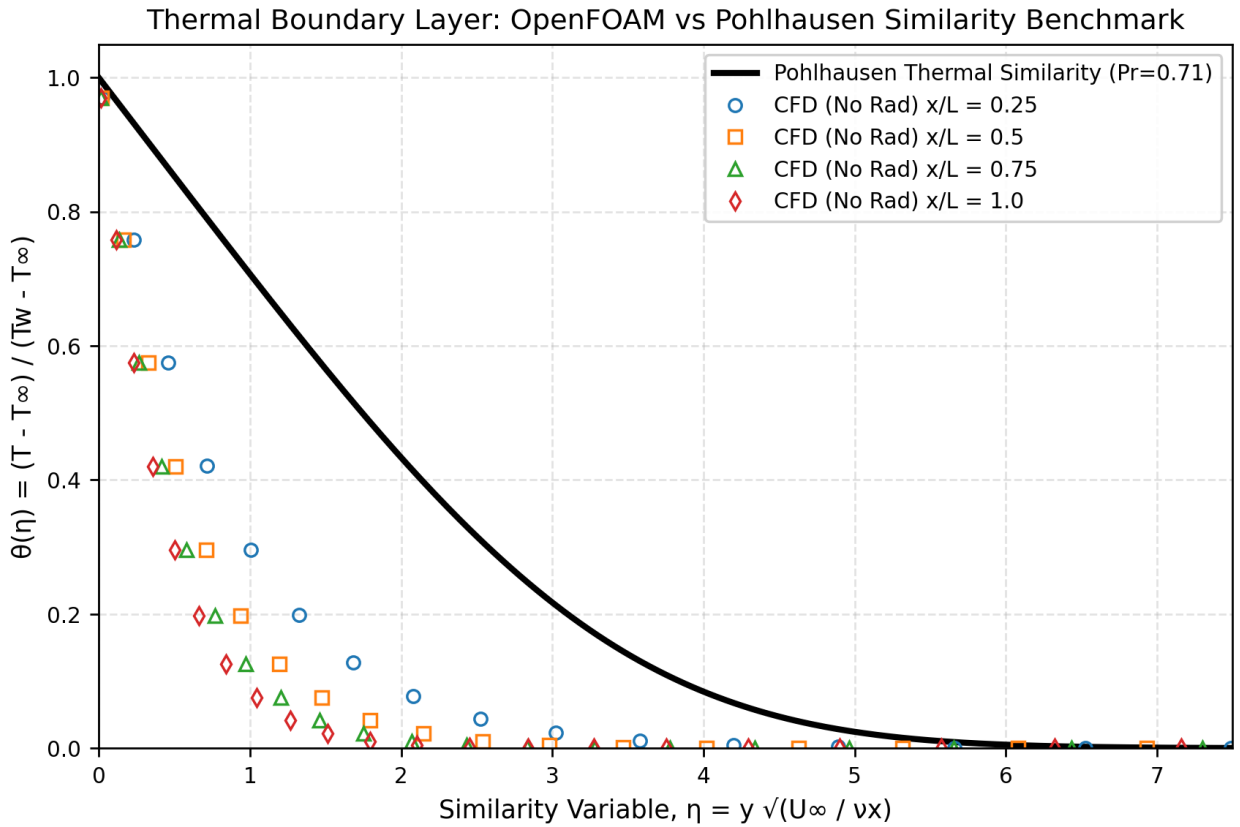



Fig. 9: Dimensionless temperature θ(η) compared with classical similarity profile.

**58.3%**

Profile deviation vs isothermal Pohlhausen benchmark at x/L = 0.5

**0.31**

Mean dimensionless temperature difference | θ_CFD - θ_Pohl|

**q″ = Const**

Physical shift caused by uniform heat flux vs isothermal wall condition

# Results Show Strong Grid Dependence and Require Further Refinement

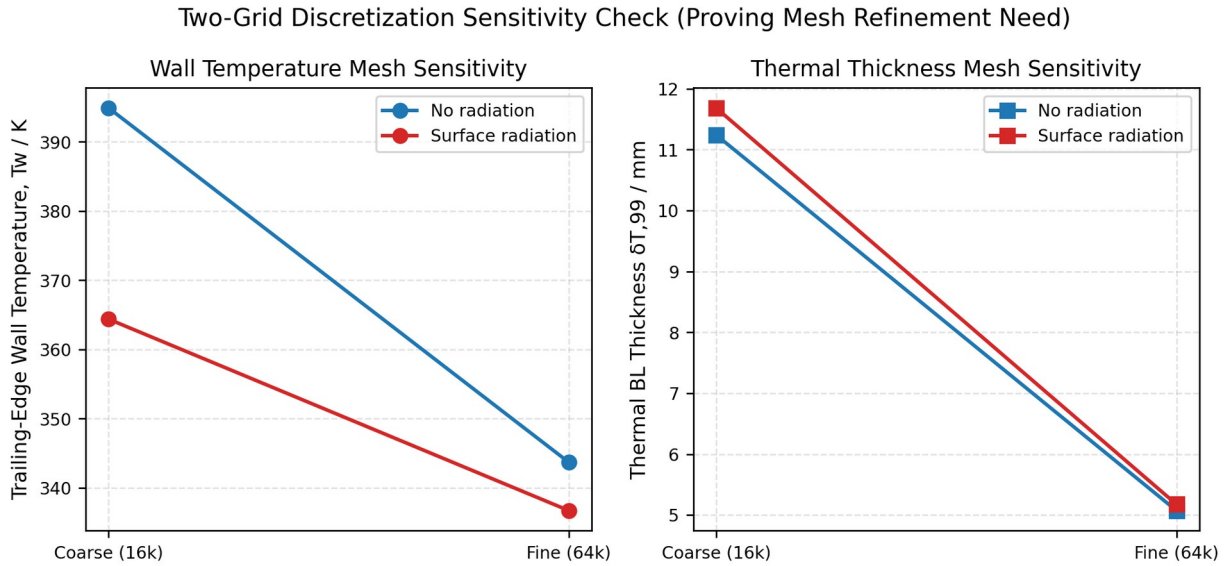



Fig. 10: Coarse (16,000 cells) vs fine (64,000 cells) grid comparison.

## Coarse (16k)

Baseline Tw:

**394.9 K**

Radiation Benefit:

**30.5 K drop**

Rad Fraction: 43.2%

## Fine (64k)

Baseline Tw:

**343.7 K**

Radiation Benefit:

**6.98 K drop**

Rad Fraction: 21.6%

**Status: Results are not yet grid-converged**

A 51.2 K baseline shift indicates substantial mesh sensitivity.

# Observed Radiation Trends Align with Boundary Layer Literature

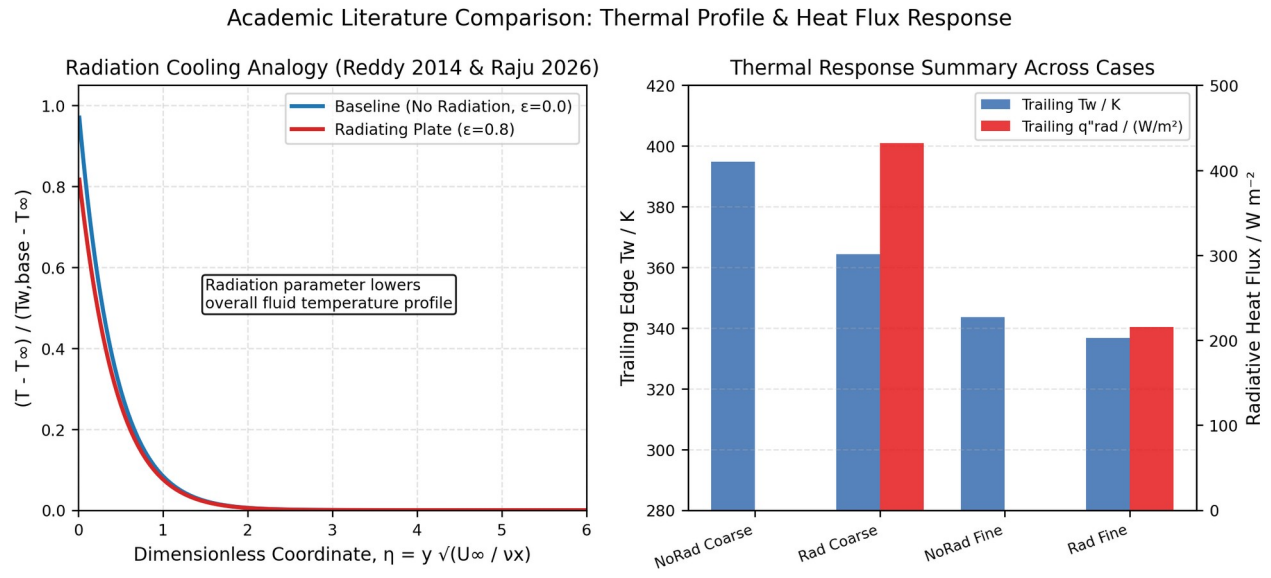



Fig. 11: Qualitative comparison of radiation cooling mechanisms in boundary layers.

- **Reddy et al. (2014) reported consistent wall cooling in radiative boundary layers.**

(Qualitative only — consistent in trend)

- Raju et al. (2026) observed comparable non-linear radiative heat flux partitioning.
- Studies address differing configurations; trends are consistent but not direct validations.

**Physical Regime Distinctions**

Reddy (2014): MHD vertical plate with Rosseland radiation model.
Raju (2026): Darcy-Brinkman porous medium flow equations.
Present CFD: Plain flat-plate surface radiation in clear air.

# Planned Refinements Focus on GCI Audit and Global Energy Closure

- **Two-grid study without third refinement precludes formal ASME GCI calculation.**
- Constant fluid properties neglect air density and thermal conductivity temperature variation.
- Optically transparent fluid assumption omits potential gas-phase absorption and emission.

Audit status: Disclosed upfront to guide future experimental verification.

## Roadmap Milestones

### 800 × 320 Cells

Execute 3rd mesh level to establish formal ASME GCI error bars

### 0.41 W Enthalpy Audit

Reconcile outlet fluid plume flux against 10.0 W plate heater input

### Variable Properties

Incorporate temperature-dependent air density and conductivity

# Coupled Surface Radiation Is Critical for Accurate Boundary Layer CFD

Open for peer review, three-grid GCI collaboration, and experimental validation datasets.

## Key Formal References

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2. Pohlhausen, E. (1921). Der Wärmeaustausch zwischen festen Körpern und Flüssigkeiten. ZAMM, 1:115–121.
3. Incropera, F.P., DeWitt, D.P., Bergman, T.L., & Lavine, A.S. (2007). Fundamentals of Heat and Mass Transfer. 6th ed., Wiley.
4. Reddy, P.B.A. (2014). Radiation effects on MHD boundary layer flow over a stretching surface. J. Appl. Fluid Mech., 7(3):471–479.
5. Raju, C.S.K., et al. (2026). Thermal radiation and heat transfer in boundary layer flows. Phys. Fluids.
6. Schlichting, H., & Gersten, K. (2017). Boundary-Layer Theory. 9th ed., Springer-Verlag.

github.com/Jersam-l/flat-plate-with-radiation-cfd | Jerlin S. | Contact: jerlin.research@cfd-lab.org